

Assignment 1: Statistical Models for Categorical Data

ADVANCED STATISTICAL LEARNING AND MODELING - MODULE B / Giulia Vannucci

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Instructions

This assignment consists of three exercises involving the application of statistical models for categorical data. Students are free to use either R or Python to complete the tasks.

Assignments may be completed individually or in groups. However, each student must submit their own version of the assignment by uploading it to the following directory on Microsoft Teams:

Assignments → Assignment1 → [create a folder named: name_surname_assignment1]

Each student's folder must contain:

- A PDF report including both exercises;
- The code used for the analysis (in '.R', '.py', or '.ipynb' format);
- A note stating whether the work was done individually or as part of a group, and if in a group, the names of all group members.

Deadline: the day of the exam. While there is no penalty for late submission, students are strongly encouraged to complete the assignment as early as possible to allow time for feedback and clarification.

Exercise 1: Logistic Modeling on Drug Consumption

Dataset: UCI Drug Consumption (Quantified)

A detailed description of all variables is available at the dataset link above.

Objective

The goal of this exercise is to investigate the effect of individual characteristics on the consumption of alcohol and cannabis, using both binary and multinomial logistic regression models.

Instructions

1. Data Preprocessing
 - Download and inspect the dataset.
 - Assign appropriate column names as described on the dataset page.
 - Prepare the data for modeling:
 - Clean the dataset;
 - Encode the target variables (alcohol and cannabis) appropriately for both binary and multinomial models.
2. Modeling

- For both substances separately (alcohol and cannabis), run a binary logistic regression model and a multinomial logistic regression model .
 - Include the following explanatory variables: Age, Gender, Education, Country, Ethnicity, Nscore, Escore, Oscore, Ascore, Cscore, Impulsiveness, Sensation_Seeking
3. Report Include in your report:
- Preprocessing steps and key decisions;
 - Interpretation and significance of model coefficients;
 - Inference on model parameters;
 - Model selection process and performance metrics;
 - Justification of modeling choices and discussion of results.
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Exercise 2: Log-Linear and Logistic Models for Acute Inflammations

Dataset : UCI Acute Inflammations Dataset

A detailed description of all variables is available at the dataset link above.

Objective

The goal of this exercise is to study associations among categorical symptoms using log-linear models , and to compare these with results from logistic regression models.

Instructions

1. Data Preparation
 - Import and inspect the dataset.
 - Ensure that categorical variables are correctly encoded (as factors or categorical types).
 2. Log-Linear Modeling
 - Analyze the relationships between: Bladder inflammation, Nausea, and Lumbar pain
 - Fit and compare different log-linear models.
 - Interpret parameter estimates and evaluate model fit.
 3. Logistic Modeling
 - Fit a logistic regression model for Bladder inflammation as a function of Nausea, Lumbar pain, and Temperature.
 4. Equivalence Between Models
 - Discuss under what conditions log-linear and logistic models are equivalent.
 - Demonstrate such equivalence using part of your analysis, where applicable.
 5. Report

Include in your report:

 - Model structures and justification;
 - Interpretation of estimated parameters;
 - Inference and hypothesis testing;
 - Model selection and comparison;
 - Discussion of equivalence between modeling approaches.
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Exercise 3: Modeling Count Data with Overdispersion

Dataset : `asthma.csv` (provided)

The dataset contains data on the number of asthmatic attacks per year among a sample of 120 patients and the associated factors.

Objective

The goal of this exercise is to model count data related to asthma episodes using Poisson and alternative models that account for overdispersion.

Instructions

1. Data Exploration
 - Load and inspect the `asthma.csv` dataset.
 - Provide a descriptive summary of the variables.
 - Create exploratory plots to examine the distribution of the count variable and potential predictors.
 2. Poisson Modeling
 - Fit a Poisson regression model to the count response variable using appropriate predictors.
 - Interpret the model coefficients and assess the goodness of fit.
 3. Check for Overdispersion
 - Compute and report diagnostic statistics.
 - Assess whether the assumption $\text{Var}(Y) = E(Y)$ is violated.
 4. Alternative Models
 - If overdispersion is present, fit alternative models.
 - Compare the models based on fit statistics and interpretability.
 5. Model Interpretation and Discussion
 - For each model, interpret relevant parameters in context.
 - Discuss the implications of overdispersion and motivate your choice of the final model.
 - Provide model predictions if appropriate, and comment on their plausibility.
 6. Report Include in your report:
 - Description of exploratory analysis and model assumptions;
 - Diagnostic checks and justification for modeling choices;
 - Interpretation of parameter estimates and inference;
 - Comparative discussion of Poisson vs. alternative models;
 - Graphical and/or tabular summary of findings.
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Submission Checklist

In your personal folder on Teams, include:

- `assignment1_report.pdf` (including all three exercises);
- The script or notebook used (`.R`, `.py`, or `.ipynb`);
- A note specifying individual or group work and, if in a group, the names of all members.

Folder structure:

Assignments → Assignment1 → name_surname_assignment1

Good luck, and don't hesitate to ask questions as you work through the assignments!